

StegaPoison: A Steganography-Inspired Untargeted Attack on Federated Recommender Systems

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Abstract

Federated recommender systems (FedRecs) are increasingly deployed in real-world platforms to enable privacy-preserving personalization. However, their distributed learning structure exposes new, largely unexplored attack surfaces. In this work, we present StegaPoison, the first steganography-based untargeted attack against federated recommendation systems. Unlike prior poisoning methods that rely on large, easily detectable perturbations, StegaPoison embeds malicious gradients using concealment-oriented information-hiding techniques. Our attack integrates five coordinated mechanisms: (1) digital watermarking, injecting structured frequency-domain signals; (2) Low-Variance Dimension Embedding Perturbation (LVDEP) inspired by Pixel-Value-Differencing (PVD) in steganography; (3) mirror-shift amplification, inducing user-item representation misalignment; (4) velocity-based sampling, which targets high-momentum parameters during training; and (5) statistical invisibility constraints to evade norm- and distribution-based defenses. Across all settings, StegaPoison achieves 94.11% to 99.88% for HR@5 degradation and 94.7% to 99.91% for NDCG@5 degradation across all datasets (MovieLens, Gowalla) and models (MF, SASRec). Our findings reveal that steganographic poisoning can substantially disrupt ranking quality while remaining statistically indistinguishable from benign client updates. StegaPoison significantly outperforms existing poisoning baselines, highlighting the urgent need for new defense mechanisms tailored to information-hiding attacks. This work contributes the first comprehensive framework for evaluating FedRec robustness against steganographic threats and establishes a new research direction at the intersection of recommender security, federated learning, and digital steganography.

Keywords: Federated Learning, Recommender Systems, Data Poisoning, Steganography, Adversarial Machine Learning, Model Robustness

1. Introduction

One promising solution to deliver tailored privacy-preserving services across distributed platforms, such as e-commerce platforms, is through Federated Recommender Systems (FedRec) Park et al. (2026). FedRec is a term used to describe decentralized recommendation systems, which is the result of integrating federated learning with recommendation (Lin et al., 2021). It alleviates growing concerns about data ownership, confidentiality, and regulatory compliance by enabling clients to work together to train recommendation models without directly sharing raw user data. Randomly chosen users train a recommendation model locally throughout each training round. They then communicate their parameter gradients to an aggregator, which aggregates all the gradients to update the parameters of the global model (Hao et al., 2024). However, the distributed nature of FedRec creates new attack surfaces in spite of these privacy guarantees, which can be broadly divided into targeted and untargeted attacks. Using methods such as poisoning or model inversion, targeted attacks seek to skew the recommendation outcome in favor of particular users or items. Untargeted attacks aim to degrade the overall performance of the model without concentrating on particular users or items. They typically use techniques such as label flipping, gradient perturbation, or data poisoning on a global level.

Although FedRec enables privacy-preserving personalization, it remains vulnerable to poisoning attacks, either targeted or untargeted. The focus of this study is mainly on untargeted attacks. Previous work mainly explores different pathways in this regard, including (1) hard sampling in item embeddings (Hao et al., 2024; Wu et al., 2022), clustering item embeddings (Yu et al., 2023), perturbation of item similarity, user model and malicious client’s sample sizes (Yi et al., 2023), and virtual user embedding sampling (Hao et al., 2024). Even though these techniques have the potential to seriously impair model convergence or recommendation accuracy, they frequently display observable statistical abnormalities such as outlier gradients, irregular loss patterns, or odd embedding norms. Consequently, federated learning pipelines with robust aggregation mechanisms, anomaly detection, or differential privacy requirements help to mitigate many of these attacks.

To get over these limitations, we propose StegaPoison—a novel, untargeted poisoning attack based on steganography principles, notably digital watermarking and LVDEP inspired by PVD. StegaPoison carefully conceals adversarial signals within updates that appear to be benign, unlike other methods that depend on overt or statistically prominent changes. Our strategy encodes disruptive gradients as low-magnitude, structured perturbations within user or item embeddings, drawing

comparisons to steganographic techniques used to conceal information within digital media. The purpose of these perturbations is to avoid conventional anomaly-based defenses by keeping them inside the statistical boundaries of typical user behavior.

The watermarking element in StegaPoison adds a structured, repeated signal to specific update vectors of user embeddings, introducing noise that mimics the patterns of natural model updates while continuously confusing the learning process. Simultaneously, by identifying pairs of dimensions with little difference and gently altering their values to express adversarial perturbations, the LVDEP module in StegaPoison takes use of the local smoothness of user embeddings. By introducing subtle signal shifts while maintaining the general structure and statistical characteristics of the embeddings, this inter-dimensional manipulation makes the poisoned update coherent and imperceptible to conventional filtering techniques. In order to develop a hybrid poisoning signal that is both effective and challenging to detect, StegaPoison combines the spread influence of watermarking with the localized stealth of LVDEP.

Moreover, a mirror shift to item embeddings is applied to escalate the misalignment between user and item vectors. This is a covert attack trick where the attacker intentionally misaligns user and item embeddings in embedding space to make them less compatible, which causes a drop in recommendation accuracy across the board. This technique is effective because it can avoid anomaly detection and cause systemic changes through tiny, regular adjustments. Instead of attacking any specific item, this results in an untargeted degradation of the model.

To further amplify the effect, velocity-based sampling is adopted, through which user-item pairs are chosen whose prediction scores change quickly over time. Optimization methods are highly effective for this purpose, as they can be utilized to identify user-item pairs with high score "velocity," adaptively update model parameters to prioritize these fast-changing relationships, and ultimately train the model more efficiently by focusing on the data that drives significant learning. Gradient-aware optimization methods are the most suitable methods that can leverage or adapt to change rates in prediction scores. Among those optimization methods, Adam optimizer is considered the best fit, as it naturally adapts to fast-changing gradients.

Finally, to ensure the stealthiness of the attack, StegaPoison adopts norm clipping as a statistical method that works against norm- and distribution-based defense mechanisms. Fig. 1 illustrates the general threat posed by untargeted poisoning attacks such as StegaPoison in federated recommender systems, where malicious clients inject carefully crafted updates that remain statistically similar to benign contributions while progressively degrading the quality of the aggregated global recommendation model.

Key Contributions

- **A Novel Stealth-Oriented Poisoning Paradigm for Federated Recommender Systems:** We introduce StegaPoi-

son, a fundamentally new untargeted attack framework that shifts the design philosophy from anomaly evasion to statistical indistinguishability, enabling adversarial updates to seamlessly blend with benign client behavior while degrading global model performance.

- **Embedding-Level Attack Modeling Beyond Global Gradient Manipulation:** We address the limitation of prior attacks that treat updates as monolithic vectors by proposing a fine-grained, embedding-aware manipulation strategy that exploits local smoothness and inter-dimensional relationships, thereby achieving high stealth against conventional distance- and similarity-based defenses.
- **Systematic Degradation via Implicit User-Item Misalignment:** Unlike existing untargeted attacks that rely on structural distortion or clustering effects, we propose a more subtle degradation mechanism that induces global semantic misalignment in the embedding space, preserving statistical properties while consistently reducing recommendation quality.
- **Adaptive Exploitation of Learning Dynamics for Attack Amplification:** We overcome the limitations of static or heuristic-based poisoning strategies by introducing an adaptive mechanism that leverages temporal sensitivity in model updates, enabling the attack to dynamically focus on high-impact interactions and maximize disruption efficiency.
- **Robust Evasion of Statistical and Robust Aggregation Defenses:** We demonstrate that carefully constrained and structured perturbations can systematically bypass widely used defense mechanisms (e.g., norm-based filtering and Byzantine-robust aggregation), highlighting critical blind spots in current defense paradigms.
- **Comprehensive Empirical Validation in Realistic Federated Settings:** Through extensive experiments on benchmark datasets (i.e., MovieLens and Gowalla), we show that the proposed approach achieves significant performance degradation while maintaining high stealth, thereby exposing the vulnerability of state-of-the-art FedRec systems under realistic conditions.

The remainder of this paper is organized as follows. Section 2 presents the background and fundamental concepts related to federated recommender systems and model poisoning attacks. Section 3 reviews existing literature on poisoning attacks in federated learning and recommendation settings. Section 4 defines the threat model and adversarial assumptions considered in this study. Section 5 introduces the proposed StegaPoison framework and describes its key components and attack strategies in detail. Section 6 reports the experimental setup and evaluation results, including attack effectiveness, robustness against existing defenses, stealthiness analysis, and ablation studies. Section 7 discusses the limitations of the current work and outlines promising directions for future research. Finally, Section 8 con-

The Problem: Stealthy Model Poisoning in Federated Recommender Systems

Malicious clients subtly manipulate their model updates so that the aggregated global model looks normal *but causes poor recommendations for all users*.

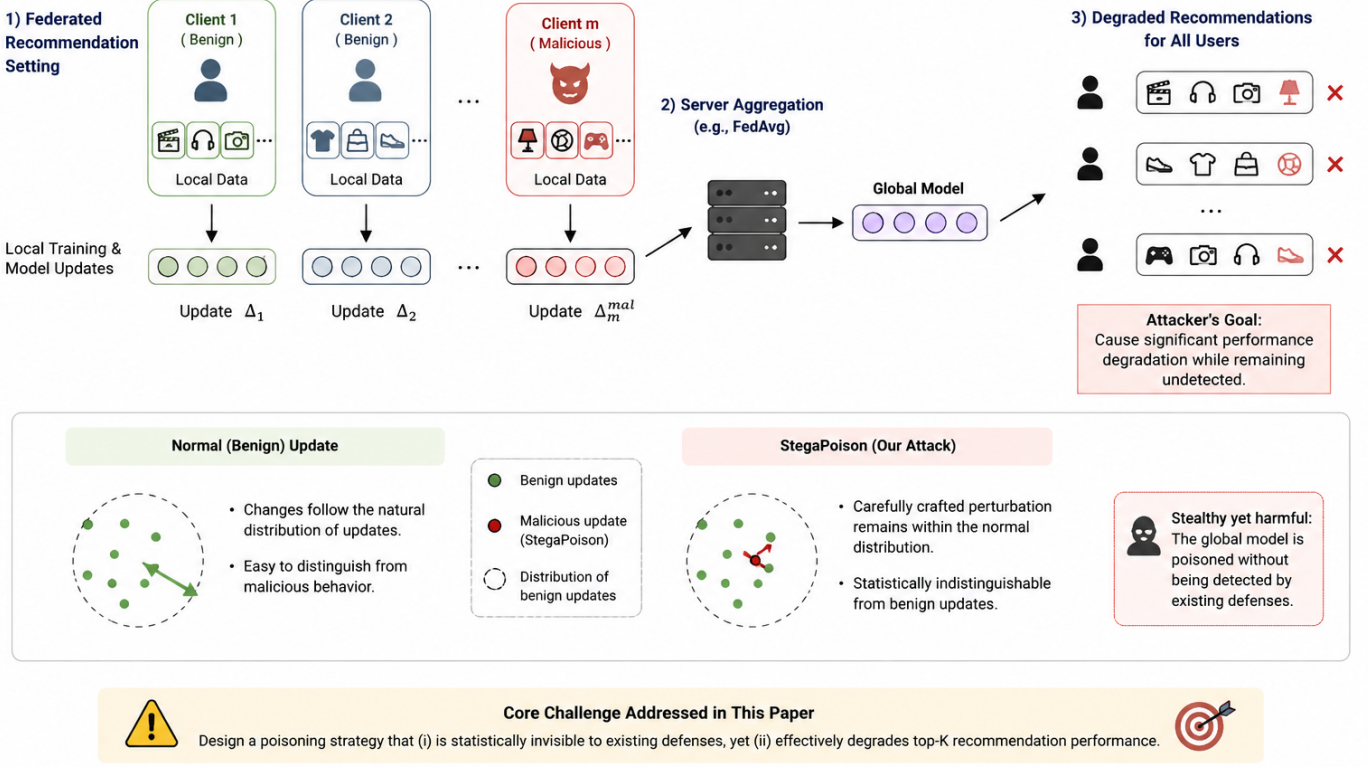


Figure 1: Stealthy Untargeted Attack in Federated Recommender Systems

cludes the paper and summarizes the main findings and contributions.

2. Background

2.1 Federated Recommender Systems

Federated learning (FL) enables model training across distributed users while keeping raw data local (McMahan et al., 2017). Federated recommender systems (FedRec) extend this paradigm to collaborative filtering models (Ammad-Ud-Din et al., 2019) such as matrix factorization (MF), neural collaborative filtering (NCF), and embedding-based architectures (Chai et al., 2021; Lin et al., 2021). In a typical FedRec system, each user maintains a personal dataset of interactions, while the server coordinates training through iterative communication rounds (Ammad-Ud-Din et al., 2019).

At training round t , the server broadcasts the global model parameters $\Theta_t = \{U_t, V_t\}$, where:

- $U_t \in \mathbb{R}^{N \times d}$ are the user embeddings,
- $V_t \in \mathbb{R}^{M \times d}$ are the item embeddings,
- d is the embedding dimensionality.

Each client updates its local user embeddings using its private interaction data and returns a model update Δ_i to the server. The server aggregates all updates via algorithms such as FedAvg (McMahan et al., 2017):

$$\Theta_{t+1} = \Theta_t + \eta \sum_{i=1}^K \frac{n_i}{n_{\text{tot}}} \Delta_i, \quad (1)$$

where n_i is the number of local interactions on client i and n_{tot} is the total across all selected clients.

2.2 Poisoning Attacks in Federated Learning

Federated learning is vulnerable to *model poisoning attacks*, in which malicious clients purposely manipulate their updates to corrupt the global model (Bhagoji et al., 2019). According to (Fang et al., 2020) these attacks can be broadly classified into **Targeted attacks** which promote/demote specific items or labels and **Untargeted attacks** that degrade the overall model performance without a specific target.

Existing attacks on FedRec typically rely on explicit gradient manipulation, amplified updates, or user embedding reconstruction based on sampling (Rong et al., 2022). However, many such strategies are detectable through norm clipping, update statistics, or clustering-based anomaly detectors (Sun et al., 2019).

2.3 Embedding-Space Vulnerabilities

Recommender systems rely heavily on embedding representations. User and item vectors are learned so that:

$$\hat{y}_{ui} = \langle U_u, V_i \rangle \quad (2)$$

captures user–item affinity. Small but structured perturbations in the embedding space can therefore create large distortions in the learned collaborative relationships (He et al., 2017).

Important vulnerability aspects include: **Low-variance dimensions**: dimensions that naturally change little across clients are ideal for hiding perturbations (Bhagoji et al., 2019), **Shift sensitivity**: user–item misalignment can significantly influence ranking metrics (Christakopoulou and Banerjee, 2019), **High-velocity parameters**: parameters with fast temporal changes represent active learning areas and are highly sensitive to corruption (Hao et al., 2024). These properties make embedding-based FedRec a high-value target for adversarial manipulation.

2.4 Steganography and Watermarking Inspiration

Steganography hides information within benign-looking signals by exploiting low-variance regions or imperceptible transformations (Barni, 2011). According to (Honsinger, 2002) digital watermarking adds structured perturbations—often sinusoidal or frequency-domain patterns—designed to be imperceptible, robust, and hard to statistically detect.

The central idea influencing this work is that **embedding spaces behave similarly to images**: certain dimensions tolerate larger changes, and structured perturbations can remain undetected while still altering model behavior (Zhang et al., 2024b). To our knowledge, no prior work applies sinusoidal / frequency-modulated watermarking directly to user/item embedding vectors in recommender systems or federated settings — StegaPoison is the first to do so.

2.5 Motivation for StegaPoison

Most existing FedRec attacks modify embeddings directly but leave detectable footprints. In contrast, StegaPoison introduces a new strategy:

- hiding perturbations in low-variance embedding dimensions,
- embedding sinusoidal watermark patterns into gradient directions,
- amplifying user–item misalignment through mirror-shift transformations,
- targeting high-velocity parameters during training,
- and ensuring statistical invisibility via norm and distribution constraints.

This combination produces a stealthy, structured, and highly effective untargeted poisoning attack that is resistant to common FL defenses.

3. Related Work

This section reviews the existing literature on untargeted poisoning attacks in federated learning and federated recommender systems, as well as the defense mechanisms developed to mitigate such threats. To provide a structured understanding of the field, the discussion is organized according to the evolutionary progression of attack methodologies, tracing the transition from early statistically detectable perturbation strategies to increasingly stealthy and adaptive poisoning techniques. We first examine the historical development of untargeted poisoning attacks, beginning with foundational statistical approaches, followed by optimization-driven attacks, and subsequently the emergence of advanced stealth-oriented manipulation strategies. The review then focuses on attacks specifically designed for federated recommender systems and analyzes the strengths and limitations of existing defense mechanisms. Building upon this analysis, we position StegaPoison within the broader attack landscape by highlighting its conceptual differences from prior approaches. In particular, we discuss how StegaPoison introduces a paradigm shift from conventional evasion-based poisoning toward statistical invisibility through steganographic embedding manipulation, inter-dimensional perturbation strategies, systemic user–item misalignment, adaptive poisoning efficiency, and coherent hybrid attack design. This progression not only illustrates the increasing sophistication of poisoning attacks but also exposes critical limitations in current defense mechanisms that motivate the development of StegaPoison.

The Evolutionary Trajectory of Untargeted Poisoning Attacks

The history of untargeted model poisoning is defined by a continuous escalation in complexity, with attackers systematically advancing their methods to circumvent progressively stronger defenses. This evolution can be segmented into three principal phases: initial strategies characterized by statistical simplicity, optimization-based attacks that introduced strategic concealment, and advanced vectors that leverage surgical precision (Lianga et al., 2023).

Foundational Strategies: The Era of Statistical Naivety

Initial attempts to compromise federated systems were largely adaptations of established vulnerabilities from centralized machine learning, focusing on two main vectors: data poisoning and model poisoning (Xia et al., 2023). Data poisoning involves manipulating local training data on compromised clients to indirectly corrupt the model updates transmitted to the server. A classic illustration is the Label-flipping attack, where an adversary reverses the labels of training instances, for example, by reassigning a label ℓ to $L - 1 - \ell$ in an L -class classification task. However, the indirect nature of this approach has proven to have a limited impact in federated environments. Malicious influence is often diluted during the aggregation of numerous legitimate updates, rendering it ineffective against systems employing even rudimentary robust aggregation protocols (Xia et al., 2023; Sikandar et al., 2023). Model poisoning, in which the adversary directly alters model updates—such as gradients or pa-

parameters—prior to transmission, was identified as a more direct and formidable threat. However, early, unsophisticated implementations of model poisoning had a significant weakness: they were statistically conspicuous. The Gaussian attack, for example, involved generating local model parameters by sampling from a Gaussian distribution (Liang et al., 2023). Such arbitrarily and unstructured modifications result in substantial statistical deviations from the updates provided by honest clients. As a result, they are readily flagged as outliers and neutralized by foundational Byzantine-robust defenses like Trimmed Mean or Krum, which are specifically engineered to reject statistically anomalous contributions. This first phase made it clear that any viable attack must be able to bypass statistical anomaly detection (Xia et al., 2023).

Optimization-Based Attacks: The Dawn of Strategic Poisoning

A pivotal moment in the development of untargeted attacks was the transition from injecting simple noise to deploying strategic, optimization-based frameworks (Xia et al., 2023). This new generation of attacks was founded on the critical insight that an update’s direction could be malicious, even if its magnitude was not unusual. The foundational research by (Fang et al., 2020) introduced the idea of a "directed deviation" attack, which casts the adversarial goal as a formal optimization problem. The objective was to construct a malicious local model that, upon aggregation, would steer the global model as far as possible in the opposite direction of its anticipated benign learning trajectory. This tactic proved remarkably successful in diminishing model accuracy while evading the statistical filtering logic of major defenses such as Krum (Blanchard et al., 2017a), Trimmed Mean, and Median (Yin et al., 2018), which primarily evaluate the magnitude or distance of updates rather than their semantic orientation. Expanding on this work, later research introduced more generalized frameworks for model poisoning. Min-Max and Min-Sum attacks, for example, defined the attacker’s goal as maximizing the malicious effect while explicitly staying within the detection thresholds of robust aggregators (Shejwalkar and Houmansadr, 2021). The Min-Max attack generates a malicious update in which the maximum distance to any benign update is constrained to be smaller than the maximum distance between any two benign updates. This method effectively operates at the edge of what a distance-based defense such as Krum would deem normal, thus avoiding detection. These optimization-based strategies represented a major advance in adversarial capabilities, proving that attackers could precisely model and circumvent specific defensive systems.

Advanced Stealth Vectors: The Move Towards Surgical Manipulation

The latest wave of general-purpose attacks has further honed this strategic methodology, emphasizing stealth and efficiency through highly focused and precise manipulations. These techniques go beyond mere evasion to minimize their statistical presence while maximizing their disruptive potential. The FedIMP attack (Li et al., 2024), for instance, leverages the concept of parameter importance. Rather than corrupting all model

parameters without distinction, FedIMP employs Fisher information to pinpoint the subset of parameters most vital to the model’s function and poisons only these high-value parameters. This precise approach improves both stealth, by reducing the overall scale of the modification, and effectiveness, by focusing the malicious payload where it will inflict the most damage. Other advanced attacks have questioned the threat model itself. The MPAF (Model Poisoning Attack based on Fake Clients) attack departs from the often-impractical scenario of compromising a significant portion of real clients. Instead, it assumes an attacker can introduce low-cost, easily simulated "fake clients" into the network. These fake clients, which hold no actual local data, work to pull the global model toward a predefined, low-performing "base model." This strategy reveals a critical and practical vulnerability by substantially lowering the resource requirements for malicious actors (Cao and Gong, 2022). Further refining the adversary’s goals, the FMFA (Flexible Model Poisoning Attack) introduced the innovative objective of fine-grained controllable attacks. Instead of a blunt denial-of-service that triggers a total model failure, FMFA allows an attacker to degrade the global model’s accuracy to a specific, predetermined level (Zhang et al., 2024a). This renders the attack significantly more covert, as a small, controlled dip in performance is far less likely to trigger alarms. This evolution from brute-force degradation to subtle, often economically driven manipulation highlights a distinct trend. The central philosophy of these attacks, however, continues to be one of evasion: creating a malicious update that is an outlier but one that is sufficiently small or cleverly positioned to bypass defensive filters. This logically raises the question of whether an attack could be engineered to produce no detectable anomaly whatsoever, a question that underpins our research.

Domain-Specific Attacks on Federated Recommender Systems

The distinct architecture of Federated Recommender Systems (FedRecs), which are built upon the interaction between user and item embeddings, has fostered a specialized category of untargeted attacks. These methods exploit the fundamental logic of recommendation to degrade performance in ways that are frequently more potent and less detectable than generic approaches. Several studies have concentrated on manipulating the training procedure or the embedding space itself. FedAttack repurposes the legitimate training method of "hard sampling" for malicious intent. It poisons the training by identifying and presenting the model with "globally hardest" samples—negative user-item pairs that are very similar and positive pairs that are very dissimilar. Submitting gradients from these confusing examples is known to impede model convergence (Wu et al., 2022). ClusterAttack is a model poisoning technique that directly targets the item embedding space, generating poisonous gradients that cause the embeddings of unrelated items to collapse into a few dense clusters. This action renders items within a cluster semantically identical to the model, thus nullifying its capacity to produce relevant rankings (Yu et al., 2023). In a similar vein, UA-FedRec executes a multi-faceted attack on federated news recommendation by concurrently perturbing

news similarity, the user model, and the reported size of local data to magnify the influence of its malicious updates (Yi et al., 2023). More recent attacks exhibit a more refined grasp of the underlying mechanics of recommendation. FRecAttack introduces a more sophisticated temporal element by targeting items with high "score velocity"—the rate at which recommendation scores change during training rounds (Hao et al., 2024). The attack recognizes that items with rapidly changing scores are more disruptive to the learning process, particularly in the initial stages. By focusing on these high-velocity items, the attack achieves greater efficacy than methods that depend solely on static properties of user-item interactions. Finally, FedRecHPA showcases the adaptability of modern adversaries by launching a hybrid attack that merges a targeted component for promoting specific items with an untargeted component to degrade overall system performance (Wang and Liu, 2024). The development of these FedRec-specific attacks demonstrates increasing sophistication in understanding and exploiting the core tenets of recommendation. While initial attacks involved relatively crude manipulations of the embedding space, subsequent attacks target more subtle concepts like learning dynamics and the interplay between user and item profiles. This indicates that the most effective future attacks will not merely inflict damage but will do so by subtly corrupting fundamental principles, such as user-item compatibility, in a manner that aligns with the learning process itself. This provides the immediate context for the novel mechanisms utilized in StegaPoison.

The Defensive Landscape and its Inherent Limitations

In parallel with the escalating complexity of poisoning attacks, a corresponding evolution has taken place in the creation of defensive mechanisms. These defenses have advanced from elementary statistical filtering to intricate, stateful analyses of client behavior and contributions. Nevertheless, despite their growing sophistication, they all share a common foundation: anomaly detection. The most fundamental and earliest defenses are stateless, Byzantine-robust aggregation rules, which seek to blunt the impact of malicious updates by classifying them as statistical outliers on a round-by-round basis. Distance-based methods such as Krum and Multi-Krum select the local model(s) that are closest to the primary cluster of benign updates in a high-dimensional space (Blanchard et al., 2017a). Coordinate-wise statistical methods such as Trimmed Mean and Median function by sorting the values for each model parameter, discarding a percentage of the highest and lowest values, and averaging the remainder (Yin et al., 2018). As noted earlier, these stateless methods are the very defenses that optimization-based attacks were engineered to circumvent. Acknowledging these shortcomings, the subsequent generation of defenses started to integrate memory and context, assessing client updates based on their behavior over time or their qualitative effect on the model. Stateful and contribution-aware defenses signify a shift from purely statistical anomaly detection to a more semantic evaluation of client trustworthiness. Behavioral analysis systems such as TESSERACT and attestedFL monitor clients across multiple rounds (Sharma et al., 2021; Mallah et al., 2023). TESSERACT computes a "flip-score" to establish

a reputation based on the directional consistency of a client's updates, while attestedFL tracks local model convergence and update similarity over time. Contribution estimation methods, such as ConDA, directly assess the quality of each client's update by using Shapley values to approximate its effect on model performance, thereby down-weighting updates considered to be detrimental (Liang et al., 2025). In the FedRec space, defenses like UNION have been developed to specifically thwart attacks such as ClusterAttack by enforcing a uniform distribution in the item embedding space, which makes the dense clusters formed by the attack a detectable anomaly (Yu et al., 2023). Despite their variety and complexity, this survey points to a unifying vulnerability: a fundamental reliance on anomaly detection. Whether the anomaly is defined as a statistical outlier (Krum), an inconsistent behavioral pattern (TESSERACT), a negative impact on model performance (ConDA), or a structural irregularity in the embedding space (UNION), the central premise is that malicious intent will manifest as a detectable deviation from a benign norm. This reactive stance, where defenses are perpetually designed to identify new types of anomalies, leaves a persistent research gap. It raises the critical question: what if an attack could be engineered from the outset to be non-anomalous across all these dimensions? This question serves as the direct and compelling motivation for StegaPoison.

Positioning StegaPoison: A Paradigm Shift Towards Statistical Invisibility

StegaPoison is engineered to overcome the fundamental weaknesses of anomaly-based defenses by introducing a new attack paradigm inspired by the principles of steganography and information hiding. In contrast to prior work that aims to evade detection, StegaPoison seeks to achieve statistical invisibility, embedding its malicious payload within updates that are statistically indistinguishable from benign ones. This section systematically contrasts the novel elements of StegaPoison with the most relevant techniques in existing literature to underscore its unique contributions.

From Evasion to Assimilation: The Steganographic Principle

The core contribution of StegaPoison is its philosophical break from previous attacks. While attacks like Min-Max are engineered to be outliers small enough to bypass filters, StegaPoison is designed to generate no detectable anomaly whatsoever. The objective is not evasion but assimilation. The malicious update is not an outlier to be concealed but a "stego-update" that masquerades as a legitimate "cover-update." This is accomplished by encoding disruptive gradients as low-magnitude, structured perturbations that are concealed within the natural statistical characteristics of user and item embeddings, drawing direct inspiration from steganographic methods used to hide information in digital media (Sumathi et al., 2014).

Inter-Dimensional Perturbation vs. Vector-Space Manipulation

The primary stealth mechanism of StegaPoison is fundamentally distinct from that of optimization-based attacks. StegaPoi-

son utilizes a Pixel Value Differencing (PVD)-based module, namely LVDEP, that leverages the local smoothness of user embeddings. It pinpoints pairs of dimensions with minimal value differences and subtly modifies them to encode adversarial signals (Wu and Tsai, 2003). This interdimensional manipulation preserves local embedding coherence. This is complemented by a digital watermarking component that introduces a structured, repetitive, low-magnitude signal to create noise that mimics natural model updates while consistently disrupting the learning process (Zhang et al., 2024c). The maliciousness is thus encoded in the subtle interdimensional relationships and repeating microscopic patterns within the update vector itself. This method stands in stark contrast to attacks like Min-Max, which handle the entire update vector as a single point in a high-dimensional space. Min-Max optimizes the position of this point to be near the cluster of benign points, thereby eluding distance-based defenses. By manipulating the internal structure of the vector, StegaPoison is designed to be significantly more covert against a wider array of defenses, including those based on vector norms or cosine similarity, which assess the vector as a whole and would be insensitive to these subtle internal rearrangements.

Systemic Misalignment vs. Embedding Space Collapse

To diminish recommendation quality, StegaPoison implements a Mirror Shift attack on item embeddings. This method perturbs item embeddings in a direction opposite to the corresponding user perturbations, deliberately creating a systemic user-item misalignment throughout the embedding space. This covertly reduces the compatibility between user and item vectors, leading to a gradual and steady decline in recommendation accuracy without targeting any particular items. This approach is considerably more precise and stealthy than the brute-force method of ClusterAttack, which collapses item embeddings into a few dense clusters. A collapsed embedding space constitutes a major structural anomaly that results in a non-uniform distribution, making it detectable by specialized defenses like UNION. A systemically misaligned or "rotated" space, as created by the Mirror Shift, can maintain its overall statistical properties (such as uniformity), making the degradation much more difficult to detect while being equally effective at harming the model's utility.

Adaptive Poisoning Efficiency vs. Static Heuristics

StegaPoison integrates a velocity-based sampling mechanism to enhance its impact by choosing user-item pairs whose prediction scores change rapidly over time. This idea of targeting high-velocity interactions was previously explored in FRecAttack (Hao et al., 2024). However, StegaPoison significantly refines this concept by pairing the sampling strategy with the Adam optimizer (Kingma and Ba, 2017). The Adam optimizer naturally adjusts to fast-changing gradients via its use of momentum estimates. By employing Adam, StegaPoison does not just pinpoint disruptive interactions; it actively harnesses the model's own learning dynamics to adaptively concentrate on and amplify perturbations where the model is most receptive to change. This fusion of a dynamic sampling strategy with

an adaptive optimizer constitutes a more deeply integrated and powerful poisoning approach than the static heuristic employed in FRecAttack.

Coherent Hybridization vs. Additive Functionality

StegaPoison is a hybrid attack, but its design philosophy diverges from previous hybrid models. The watermarking, LVDEP and mirror shift elements are not separate attacks but are synergistic components operating in unison toward the singular goal of stealthy, untargeted degradation. The watermarking provides diffuse confusion across multiple updates, the LVDEP ensures localized stealth within individual vectors, and the mirror shift executes the primary attack on user-item compatibility. These methods are intertwined to generate a poisoning signal that is both potent and difficult to deconstruct. This is different from an attack like FedRecHPA, which is a hybrid of two distinct goals: targeted item promotion and untargeted system degradation. StegaPoison's design is a coherent hybrid of complementary mechanisms, resulting in a more robust strategy where the different components can obscure each other's statistical signatures.

4. Threat Model

4.1 Adversarial Goal

The objective of the adversary is to perform an *untargeted poisoning attack* that degrades the overall quality of recommendations in a federated recommender system (FedRec). Instead of promoting or demoting a specific item, the attacker seeks to introduce systematic user-item *misalignment*, causing the global model to learn distorted collaborative patterns that reduce metrics such as HR@K and NDCG@K for all users (Yu et al., 2023).

4.2 Adversary Capabilities

We assume a realistic *semi-honest* federated learning environment in which:

- The attacker controls a small subset of clients, denoted by $\mathcal{A} \subset \mathcal{C}$, where $|\mathcal{A}| \ll |\mathcal{C}|$.
- Malicious clients fully participate in the FL protocol and receive the same model parameters broadcast by the server.
- Each malicious client can modify its local model updates before sending them back to the server.
- No access to other users' raw data is permitted, and the attacker does not observe benign clients' local gradients.

This is aligned with common security assumptions in federated recommendation settings, where local data remains private and each client only contributes parameter updates (Wu et al., 2022; Yu et al., 2023; Hao et al., 2024).

4.3 Adversary Knowledge

We assume that the adversary possesses:

- Knowledge of the model architecture (e.g., MF, NCF), loss function, and optimizer (Bhagoji et al., 2019).
- Knowledge of the embedding dimensionality and update format (Fang et al., 2020).
- No access to the training data distribution of benign clients (Bagdasaryan et al., 2020).
- No ability to observe server-side aggregation beyond the publicly shared global model parameters (Yu et al., 2023).

This corresponds to a *white-box model architecture* in which the adversary has knowledge of the recommender model architecture and training process, *black-box data* setting where access to the complete global training dataset is unavailable and only local client data are observable (Papernot et al., 2017).

4.4 Adversarial Manipulation Space

Malicious clients may alter only the updates they send to the server. Specifically, for a malicious client i , the attacker can replace its legitimate update Δ_i with a *poisoned* update $\tilde{\Delta}_i$:

$$\tilde{\Delta}_i = \mathcal{A}(\Delta_i, \Theta_t), \quad (3)$$

where Θ_t denotes the global model received from the server and $\mathcal{A}(\cdot)$ denotes the attacker’s manipulation function.

The attacker cannot:

- Modify server-side aggregation rules.
- Interfere with communication between benign clients and the server.
- Access or alter benign users’ local datasets.

4.5 Stealth Requirements

To remain undetected, the adversary must ensure that the poisoned update $\tilde{\Delta}_i$:

- Obeys norm-based defenses:

$$\|\tilde{\Delta}_i - \Delta_i\|_2 \leq \gamma \|\Delta_i\|_2, \quad (4)$$

for a small γ .

- Maintains statistical similarity to benign updates in terms of mean and variance.
- Avoids detectable anomalies in parameter distributions or outlier-based defenses.

StegaPoison achieves stealth by embedding perturbations in low-variance dimensions, applying sinusoidal watermarking, and using norm-bounded mirror-shift and velocity-driven perturbations.

4.6 Scope of Attack

StegaPoison is designed for **any federated recommender system** whose model includes:

- user embeddings $U \in \mathbb{R}^{N \times d}$,
- item embeddings $V \in \mathbb{R}^{M \times d}$,
- or hybrid neural architectures with shared embedding layers.

The attack is compatible with FedAvg, FedOpt, and other aggregation strategies, as it manipulates the update vector rather than the data distribution.

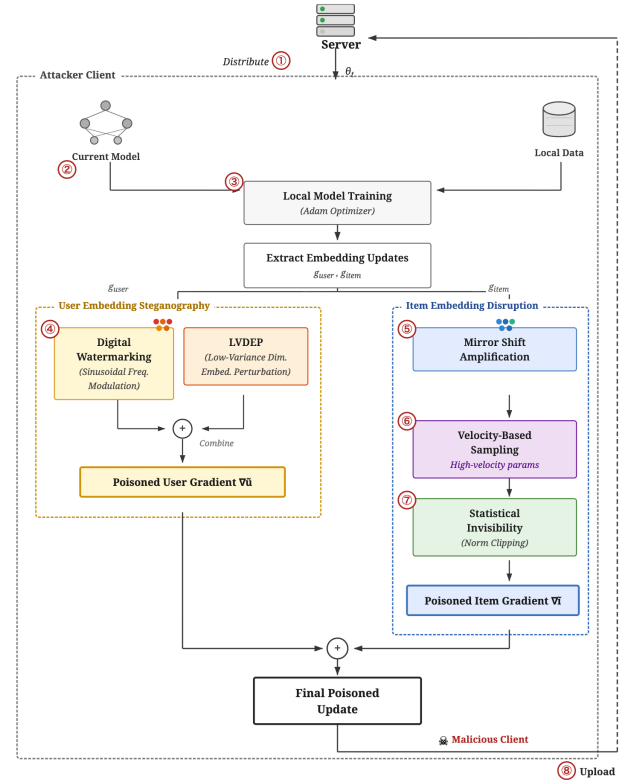


Figure 2: The framework of our StegaPoison attack

5. Proposed Method: StegaPoison

In this section, we introduce the overall workflow of **StegaPoison**, as illustrated in Fig. 2. In each communication round, the malicious client first receives the latest global model parameters from the server and performs standard local training on its private interaction data to obtain intermediate embedding gradients for the user and item. Instead of directly transmitting these benign updates, the client strategically modifies them through a coordinated multi-stage poisoning process designed to maximize degradation while preserving statistical normality. On the user-embedding side, two complementary steganography-inspired mechanisms are employed: **digital watermarking**, which injects a structured low-magnitude perturbation using sinusoidal frequency modulation, and **Low-Variance Dimension**

Embedding Perturbation (LVDEP), which selectively alters dimensions that naturally exhibit low variation, thereby concealing adversarial signals within locally stable embedding regions. These two components are then combined to generate a poisoned user gradient. In parallel, the item-embedding branch applies **Mirror Shift Amplification**, which perturbs item representations in directions opposite to the manipulated user embeddings to induce systematic user–item compatibility misalignment. To further enhance attack efficiency, **velocity-based sampling** guided by the Adam optimizer prioritizes rapidly changing user–item interactions, where perturbations are expected to exert greater downstream influence. Finally, a **statistical invisibility** stage based on norm clipping constrains the magnitude of the malicious update such that it remains difficult to distinguish from benign client contributions under common anomaly-detection rules. The poisoned user and item gradients are subsequently merged into a final malicious update and uploaded to the server, where they are aggregated with other client updates to gradually deteriorate the quality of the recommendation.

5.1 Step 1: Digital Watermarking via Sinusoidal Frequency Modulation

Let d denote the embedding dimensionality. StegaPoison generates a structured watermark vector $w \in \mathbb{R}^d$ composed of a random amplitude component and a frequency-modulated sinusoidal carrier (Honsinger, 2002). First, we draw a random base pattern:

$$a_j \sim \mathcal{N}(0, \sigma_w^2), \quad j = 1, \dots, d, \quad (5)$$

where σ_w controls the watermark strength.

Next, we define a normalized frequency axis

$$f_j = \frac{j}{d}, \quad j = 1, \dots, d, \quad (6)$$

and modulate the amplitudes using a sinusoidal carrier with frequency parameter ω :

$$w_j = a_j \cdot \sin(2\pi \omega f_j). \quad (7)$$

This yields a structured watermark that oscillates smoothly across embedding dimensions:

$$w = [w_1, w_2, \dots, w_d]. \quad (8)$$

Given a client embedding update matrix $E \in \mathbb{R}^{n \times d}$, StegaPoison injects the watermark into every updated embedding vector:

$$E_{i,j}^{(\text{wm})} = E_{i,j} + w_j, \quad i = 1, \dots, n, \quad j = 1, \dots, d. \quad (9)$$

Equivalently, in matrix form:

$$E^{(\text{wm})} = E + \mathbf{1}_n w^\top, \quad (10)$$

where $\mathbf{1}_n$ is an n -dimensional vector of ones.

The sinusoidal watermark produces smooth, frequency-domain structure that is: (i) difficult for anomaly detectors to isolate, (ii) spread across many dimensions, and (iii) preserved through federated averaging due to its consistent direction.

5.2 Step 2: Low-Variance Dimension Embedding Perturbation (LVDEP)

Let $E \in \mathbb{R}^{n \times d}$ denote the client-side embedding update matrix, where n is the number of updated user vectors and d is the embedding dimensionality. For each embedding dimension j , we compute its empirical variance:

$$\sigma_j^2 = \text{Var}(E_{:,j}). \quad (11)$$

Dimensions exhibiting low variance are identified using a pre-defined threshold τ :

$$L = \{j \mid \sigma_j^2 < \tau\}, \quad (12)$$

where L is the set of dimensions with low variance and $|L|$ denotes the concealment capacity.

Following the Pixel–Value–Differencing (PVD) principle, StegaPoison examines the difference between adjacent embedding values:

$$d_{i,j} = |E_{i,j} - E_{i+1,j}|, \quad (13)$$

and a position (i, j) is eligible for steganographic perturbation if and only if

$$j \in L \quad \text{and} \quad d_{i,j} < \tau. \quad (14)$$

For each eligible entry, a small Gaussian perturbation is injected to conceal the poisoning signal:

$$E_{i,j}^{(\text{new})} = E_{i,j} + \epsilon_{i,j}, \quad (15)$$

where

$$\epsilon_{i,j} \sim \mathcal{N}(0, \sigma_\epsilon^2), \quad (16)$$

and σ_ϵ^2 is chosen to keep the perturbation magnitude imperceptible.

Let $M \in \{0, 1\}^{n \times d}$ denote the binary mask specifying all LVDEP-eligible entries:

$$M_{i,j} = \begin{cases} 1, & \text{if } j \in L \text{ and } d_{i,j} < \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (17)$$

The final embedding update is thus given by:

$$E^{(\text{concealed})} = E + M \odot \epsilon, \quad (18)$$

where $\odot$ denotes element-wise multiplication.

We employ a single threshold τ for both variance-based dimension selection and local difference evaluation, enforcing a conservative stealth criterion that jointly captures global stability and local smoothness.

5.3 Step 3: Mirror Shift Amplification

Let $\theta^{(g)}$ denote the global model parameters broadcast by the server and let $\theta^{(c)}$ denote the client’s locally updated parameters. The *Mirror Shift* attack amplifies the natural deviation of the client update from the global model to create user–item embedding misalignment.

Shift Direction: For any parameter tensor (e.g. user or item embeddings), the natural update direction is computed as

$$\Delta\theta = \theta^{(c)} - \theta^{(g)}. \quad (19)$$

Mirror Shift Amplification: The attacker amplifies this shift by a factor $\alpha > 0$:

$$M = \alpha \cdot \Delta\theta. \quad (20)$$

Poisoned Update: The amplified shift is injected back into the client update:

$$\theta^{(\text{ms})} = \theta^{(c)} + M = \theta^{(c)} + \alpha(\theta^{(c)} - \theta^{(g)}). \quad (21)$$

Equivalently, this can be written as:

$$\theta^{(\text{ms})} = (1 + \alpha)\theta^{(c)} - \alpha\theta^{(g)}. \quad (22)$$

The Mirror Shift pushes user and item embeddings further from the global representation, creating systematic misalignment that degrades collaborative filtering quality while remaining subtle enough to evade norm-based anomaly detectors.

5.4 Step 4: Velocity-Based Sampling

Velocity-based sampling targets parameters that exhibit rapid changes across federated training rounds. These *high-velocity* parameters capture emerging patterns in user-item interactions and therefore provide the most effective leverage for poisoning.

Parameter Velocity: Let $\theta_r^{(c)}$ denote a client parameter at round r . The velocity (first-order temporal derivative) is defined as

$$v_r = \theta_r^{(c)} - \theta_{r-1}^{(c)}. \quad (23)$$

Velocity Magnitude: To assess its importance, we compute the norm:

$$\|v_r\| = \|\theta_r^{(c)} - \theta_{r-1}^{(c)}\|. \quad (24)$$

If the magnitude exceeds a predefined threshold τ_{vel} , the parameter is considered *high-velocity* and becomes a target for attack:

$$\|v_r\| > \tau_{\text{vel}} \Rightarrow \text{apply perturbation.} \quad (25)$$

Perturbation Generation: For high-velocity parameters, we generate a Gaussian perturbation scaled by the velocity:

$$\epsilon_r \sim \mathcal{N}(0, \sigma_r^2 I), \quad \sigma_r = \min(0.1\|v_r\|, \sigma_{\text{max}}). \quad (26)$$

Poisoned Update: The perturbed parameter is then:

$$\theta_r^{(\text{vel})} = \theta_r^{(c)} + \epsilon_r. \quad (27)$$

By steering perturbations toward parameters undergoing rapid learning, the attacker maximizes disruption while keeping the overall update norm small, making the attack difficult to detect.

5.5 Step 5: Statistical Invisibility

To evade norm-based and distribution-based defenses, the attack ensures that the poisoned update remains statistically indistinguishable from the benign update. This is achieved by enforcing constraints on the perturbation norm, while also preserving first- and second-order statistics such as mean and standard deviation.

Perturbation and Norm Constraint: Let $\theta^{(o)}$ denote the original (benign) update and $\theta^{(p)}$ the poisoned update. The perturbation is

$$\Delta = \theta^{(p)} - \theta^{(o)}. \quad (28)$$

The attack enforces a norm constraint:

$$\|\Delta\| \leq \alpha \|\theta^{(o)}\|, \quad (29)$$

where α is a norm-scaling constant (e.g., $\alpha = 1.2$).

If the constraint is violated, a scaling factor is applied:

$$s = \frac{\alpha \|\theta^{(o)}\|}{\|\Delta\|}, \quad \Delta^{(\text{scaled})} = s \cdot \Delta. \quad (30)$$

The norm-corrected poisoned update becomes:

$$\theta^{(p,n)} = \theta^{(o)} + \Delta^{(\text{scaled})}. \quad (31)$$

Statistical Property Preservation. To remain invisible under statistical detectors, the poisoned update must preserve the mean and standard deviation of the original update.

Let

$$\begin{aligned} \mu_o &= \text{mean}(\theta^{(o)}), & \sigma_o &= \text{std}(\theta^{(o)}), \\ \mu_p &= \text{mean}(\theta^{(p,n)}), & \sigma_p &= \text{std}(\theta^{(p,n)}). \end{aligned}$$

If the mean deviates beyond a threshold τ_{stat} :

$$|\mu_p - \mu_o| > \tau_{\text{stat}} \Rightarrow \theta^{(p,n)} \leftarrow \theta^{(p,n)} + (\mu_o - \mu_p). \quad (32)$$

If the standard deviation deviates:

$$|\sigma_p - \sigma_o| > \tau_{\text{stat}} \Rightarrow \theta^{(p,n)} \leftarrow (\theta^{(p,n)} - \mu_p) \cdot \frac{\sigma_o}{\sigma_p} + \mu_p. \quad (33)$$

By preserving both the norm and statistical moments, this technique ensures that the attack remains undetected by norm clipping, anomaly detection, and distribution-based defenses.

The individual components of StegaPoison described above operate jointly within a coordinated poisoning pipeline. Although the preceding subsections detail each mechanism separately, their sequential integration determines the final malicious update transmitted to the server. For clarity and reproducibility, the complete procedural workflow of the proposed attack, including standard local training by each client, is summarized in Algorithm 1, which outlines the generation of poisoned user and item gradients, their stealth-oriented refinement, and the construction of the final adversarial client update.

Algorithm 1 StegaPoison: Steganography-Inspired Untargeted Attack

Require: Client c , global params $\Theta^{(g)}$, local data $\mathcal{D}_c$, frequency ω , strength γ , LVDEP threshold τ_{lvdep} , mirror factor β , velocity threshold τ_{vel} , norm limit α , stat threshold τ_{stat}

- 1: **procedure** STEGAPOISON($\Theta^{(g)}, \mathcal{D}_c$)
- 2: **Step 1: Local Training**
- 3: Train on $\mathcal{D}_c$ to get $\Theta^{(c)}$
- 4: $\Delta^{(c)} \leftarrow \Theta^{(c)} - \Theta^{(g)}$
- 5: **Step 2: Watermark Injection**
- 6: Generate watermark: $w_j = \gamma \sin(2\pi\omega j)$
- 7: $\Delta^{(wm)} \leftarrow \Delta^{(c)} + w$
- 8: **Step 3: LVDEP**
- 9: Identify low-variance dims: $\mathcal{L} = \{j \mid \text{Var}_j(\Delta^{(c)}) < \tau_{lvdep}\}$
- 10: For $j \in \mathcal{L}$: $\Delta_{ij}^{(lvdep)} = \Delta_{ij}^{(wm)} + \epsilon_{ij}$
- 11: **Step 4: Mirror Shift**
- 12: $d \leftarrow \Delta^{(lvdep)}$
- 13: $\Delta^{(ms)} \leftarrow \Delta^{(lvdep)} + \beta d$
- 14: **Step 5: Velocity-Based Sampling**
- 15: $v \leftarrow \Delta^{(ms)} - \Delta_{prev}^{(c)}$
- 16: **if** $\|v\| > \tau_{vel}$ **then**
- 17: $\Delta^{(vel)} \leftarrow \Delta^{(ms)} + \eta \cdot \mathcal{N}(0, I)$
- 18: **else**
- 19: $\Delta^{(vel)} \leftarrow \Delta^{(ms)}$
- 20: **end if**
- 21: **Step 6: Statistical Invisibility**
- 22: **if** $\|\Delta^{(vel)}\| > \alpha \|\Theta^{(g)}\|$ **then**
- 23: $\Delta^{(scaled)} \leftarrow \frac{\alpha \|\Theta^{(g)}\|}{\|\Delta^{(vel)}\|} \Delta^{(vel)}$
- 24: **else**
- 25: $\Delta^{(scaled)} \leftarrow \Delta^{(vel)}$
- 26: **end if**
- 27: Match statistics to clean update:
- 28: $\text{mean}(\Delta^{(f)}) \leftarrow \text{mean}(\Delta^{(c)})$
- 29: $\text{std}(\Delta^{(f)}) \leftarrow \text{std}(\Delta^{(c)})$
- 30: **return** $\Delta^{(f)}$
- 31: **end procedure**

6. Experiments

This section evaluates the effectiveness, stealth, and robustness of STEGAPOISON across two widely used recommendation datasets, two backbone models, and multiple baselines. All experiments are implemented in a federated learning (FL) environment with partial client participation.

6.1 Experimental Setting

This subsection presents the experimental settings used to evaluate the effectiveness, robustness, and stealthiness of the proposed StegaPoison attack. Specifically, we describe the fed-

erated learning configuration, training protocol, and computational environment employed throughout the experiments. We then introduce the datasets, recommendation models, baseline attack methods, and evaluation metrics used for performance assessment. Finally, we present the defense mechanisms considered in our evaluation and discuss their role in assessing the robustness of StegaPoison against existing protection strategies.

Federated Configuration

Following the experimental protocol of (Yu et al., 2023) [ClusterAttack], we simulate a federated recommendation system where each user serves as an independent FL client. The total client population comprises 6,040 users for MovieLens-1M and 29,858 users for Gowalla. In each communication round, 50 clients are randomly sampled to participate in local training and gradient upload. Among all clients, 5% are designated as malicious (i.e., 302 attackers for ML-1M and 1,492 for Gowalla), representing a moderately stronger adversarial budget than the 1% default in [ClusterAttack] to better evaluate the steganographic attack capabilities. Even at 5%, StegaPoison achieves significantly higher degradation (94–99%) than ClusterAttack (30–41%), demonstrating its superior attack effectiveness. Each experiment is trained for 6,000 communication rounds to ensure convergence, and we report the mean results across 5 independent runs with different random seeds. The experimental hyperparameters are summarized in Table 1.

System Requirements

Experiments require minimal compute resources. **Minimum:** Python 3.8+, 8GB RAM, 5GB storage. **Recommended:** Python 3.9+, 16GB RAM, optional GPU acceleration. The results reported below were obtained on a machine with Python 3.10 and an NVIDIA GPU.

Datasets

We evaluate the proposed StegaPoison attack on two widely used benchmark datasets for top-(K) recommendation tasks, namely MovieLens-1M and Gowalla. MovieLens-1M is one of the most extensively adopted movie-rating datasets in recommender systems research and contains explicit user-item rating interactions, making it a standard benchmark for evaluating recommendation performance (Harper and Konstan, 2015). Gowalla is a location-based social networking dataset in which user check-ins are treated as implicit user-item interactions, providing a realistic setting for evaluating recommendation models under sparse and large-scale interaction data (Cho et al., 2011). Together, these datasets represent two distinct recommendation scenarios, enabling a comprehensive assessment of attack effectiveness across different data characteristics and interaction patterns. Detailed statistics of the datasets are reported in Table 2.

Models

To evaluate the generality of StegaPoison across different recommendation paradigms, we conduct experiments on two representative recommender architectures: Matrix Factorization

Table 1: Hyperparameter Settings for Different Models and Datasets

Hyperparameters	ML-1M MF	ML-1M SASRec	Gowalla MF	Gowalla SASRec
Learning rate (η)	$2e-3$	$1e-3$	$2e-3$	$1e-3$
ℓ_2 regularization (λ)	$1e-5$	$1e-5$	$1e-6$	$1e-5$
L2 norm bound (NormBound)	0.1	1.5	0.1	2.5
StegaPoison ratio	0.05	0.05	0.05	0.05
StegaPoison stealth	1.5	1.5	1.5	1.5

Table 2: Dataset statistics used in the experiments.

Dataset	# Users	# Items	# Interactions
MovieLens-1M	6,040	3,043	1,000,209
Gowalla	29,858	40,981	1,027,370

(MF) and SASRec. MF is a classical collaborative filtering model that learns latent user and item embeddings to capture preference patterns and remains one of the most widely adopted recommendation approaches in both academic research and industrial applications (Koren, 2008). In contrast, SASRec is a transformer-based sequential recommender that leverages self-attention mechanisms to model users’ historical interaction sequences and capture complex temporal dependencies among items (Kang and McAuley, 2018). By evaluating StegaPoison on both MF and SASRec, we assess its effectiveness across fundamentally different recommendation architectures, ranging from simple latent-factor embedding models to more sophisticated attention-based sequential recommendation systems.

Baselines

We compare StegaPoison with five state-of-the-art federated poisoning benchmarks:

- **LabelFlip** (Tolpegin et al., 2020): flips the label of the training data on malicious clients.
- **Gaussian** (Fang et al., 2020): uploads samples from the estimated Gaussian distribution after calculating the mean and standard deviation of normal gradients.
- **LIE** (Baruch et al., 2019): slightly distorts the average normal gradients.
- **Fang** (Fang et al., 2020): modifies the average normal gradient by adding noise in opposing directions.
- **FedAttack** (Wu et al., 2022): gradient amplification attack.
- **ClusterAttack** (Yu et al., 2023): clustering-based malicious client contribution.

These attacks represent the dominant poisoning vectors in federated learning, including FedRec, literature.

Evaluation Metrics

To assess the impact of StegaPoison on recommendation qual-

ity, we employ two widely used top-(K) evaluation metrics: Hit Rate (HR@K) and Normalized Discounted Cumulative Gain (NDCG@K). HR@K measures whether the ground-truth item appears among the top-(K) recommended items and therefore reflects the model’s ability to successfully retrieve relevant items for users (Wu et al., 2022) (Yu et al., 2023). NDCG@K, on the other hand, evaluates the quality of the ranking by assigning higher importance to relevant items appearing at higher positions in the recommendation list, thereby providing a more fine-grained assessment of recommendation effectiveness (Wu et al., 2022)(Yu et al., 2023). Following established practice in the federated recommender systems literature, we report HR@5 and NDCG@5 throughout all experiments, as these metrics effectively capture the impact of poisoning attacks on top-ranked recommendations that are most visible to end users.

Defense Mechanisms Evaluated

To assess the robustness and stealth of StegaPoison, we evaluated its performance under a comprehensive suite of state-of-the-art federated learning defenses. These defenses span classical Byzantine-resilient aggregation, norm-based filtering, and recent multi-round behavioral analysis methods.

- **Krum and MultiKrum**: Krum (Blanchard et al., 2017a) selects the single update closest to the majority of benign clients by minimizing the sum of distances to its closest neighbors. MultiKrum extends this to select multiple such updates and averages them, enabling improved stability in FL settings. Both defenses are distance-based and aim to filter statistically deviant gradients.
- **Trimmed Mean**: Trimmed Mean (Blanchard et al., 2017b) performs coordinate-wise sorting of all client updates and removes the largest and smallest values before averaging. This defense protects against outliers on a per-dimension basis, mitigating high-magnitude poisoning.
- **NormBound**: NormBound normalizes or clips each client update to a fixed ℓ_2 -norm threshold, preventing adversaries from injecting high-energy perturbations. This defense is widely deployed due to its simplicity and compatibility with FedAvg.
- **FL-WBC**: FL-WBC (Weak Byzantine Consensus) employs statistical filtering across clients to reject updates that display inconsistent directional behavior. It aims to identify anomalous update patterns rather than relying solely on norms or distances.

Table 3: Model Performance Under Different Untargeted Attack Methods (No Defense).

Model	Attack Method	ML-1M HR@5 ↓	ML-1M nDCG@5 ↓	Gowalla HR@5 ↓	Gowalla nDCG@5 ↓
MF	No Attack	0.03549 (-)	0.02226 (-)	0.02523 (-)	0.01697 (-)
	LabelFlip	0.03561 (-0.34%)	0.02238 (-0.54%)	0.02541 (-0.71%)	0.01711 (-0.82%)
	FedAttack	0.03358 (5.38%)	0.02118 (4.85%)	0.02371 (6.02%)	0.01585 (6.60%)
	Gaussian	0.03555 (-0.17%)	0.02224 (0.09%)	0.02528 (-0.20%)	0.01701 (-0.24%)
	LIE	0.03259 (8.17%)	0.02062 (7.37%)	0.02316 (8.20%)	0.01571 (7.42%)
	Fang	0.03038 (14.40%)	0.01897 (14.78%)	0.02131 (15.54%)	0.01448 (14.67%)
	ClusterAttack	0.02451 (30.94%)	0.01545 (30.59%)	0.01664 (34.05%)	0.01117 (34.18%)
	StegaPoison (Ours)	0.00209 (94.11%)	0.00123 (94.46%)	0.00014 (99.44%)	0.00010 (99.43%)
SASRec	No Attack	0.10810 (-)	0.07053 (-)	0.03251 (-)	0.02217 (-)
	LabelFlip	0.10857 (-0.43%)	0.07071 (-0.26%)	0.03270 (-0.58%)	0.02222 (-0.23%)
	FedAttack	0.10013 (7.37%)	0.06572 (6.82%)	0.03054 (6.06%)	0.02087 (5.86%)
	Gaussian	0.10769 (0.38%)	0.07055 (-0.03%)	0.03226 (0.77%)	0.02222 (-0.23%)
	LIE	0.09677 (10.48%)	0.06281 (10.95%)	0.03008 (7.47%)	0.02021 (8.84%)
	Fang	0.08964 (17.08%)	0.05909 (16.22%)	0.02797 (13.96%)	0.01883 (15.07%)
	ClusterAttack	0.06547 (39.44%)	0.04130 (41.44%)	0.02223 (31.62%)	0.01544 (30.36%)
	StegaPoison (Ours)	0.00558 (94.84%)	0.00339 (95.19%)	0.00769 (76.36%)	0.00556 (74.93%)

*Percentages in parentheses denote relative performance degradation compared with the No Attack baseline. Lower HR@5 and nDCG@5 values indicate stronger untargeted attack effectiveness.

- **Union-Based Defenses:** We additionally evaluate two hybrid defenses that combine classical and behavioral mechanisms (Yu et al., 2023):

- **MultiKrum+Union:** applies MultiKrum for initial filtering, followed by a union-based check that compares update directionality with historical benign behavior.
- **NormBound+Union:** enforces norm clipping and then incorporates multi-round consistency checks to detect suspicious convergence patterns.

Purpose of Defense Evaluation

Including these defenses allows us to rigorously test whether StegaPoison’s steganographic perturbations—designed to be low-magnitude, statistically consistent, and directionally inconspicuous—can bypass both *stateless* (Krum, MultiKrum, TrimmedMean, NormBound) and *stateful* (FL-WBC, Union-based) detection mechanisms.

6.2 Evaluation of the Attack Performance without Defense

A comparative analysis of StegaPoison with other existing untargeted attacks is presented in Table 3. The table reports the attack effectiveness of StegaPoison against several representative untargeted poisoning baselines under the no-defense setting. The evaluation is conducted using two recommender architectures, namely Matrix Factorization (MF) and SASRec, across two benchmark datasets (MovieLens-1M and Gowalla), with recommendation quality measured by HR@5 and nDCG@5. Since lower values indicate stronger attack impact, the results clearly demonstrate the superior destructive capability of StegaPoison over all competing methods.

For the MF model, StegaPoison reduces HR@5 and nDCG@5 on ML-1M by 94.11% and 94.46%, respectively, substantially outperforming the strongest baseline ClusterAttack, which achieves reductions of only 30.94% and 30.59%. On Gowalla, the degradation becomes even more severe, where StegaPoison lowers HR@5 and nDCG@5 by 99.44% and 99.43%, respectively. Similar trends are observed for SASRec. On ML-1M, StegaPoison decreases HR@5 and nDCG@5 by 94.84% and 95.19%, while the strongest baseline yields only 39.44% and 41.44% reductions. On the Gowalla dataset, although StegaPoison substantially degrades the recommendation model, the remaining recommendation quality is still comparatively higher than in other experimental settings. Nevertheless, the attack achieves significant performance reductions of 76.36% in HR@5 and 74.93% in nDCG@5, demonstrating its strong destructive capability even under more challenging recommendation dynamics.

These results indicate that StegaPoison consistently generalizes across both shallow (MF) and sequential deep recommendation models (SASRec), as well as across datasets with different interaction characteristics. The large performance gap relative to existing attacks further suggests that embedding-level stealth perturbations provide a significantly more effective mechanism for untargeted poisoning in federated recommender systems.

6.3 Attack Robustness Under Existing Defenses

Fig. 3 presents the robustness of StegaPoison under seven representative aggregation-based defense mechanisms, including TrimmedMean, Krum, MultiKrum, NormBound, FL-WBC, MultiKrum+UNION, and NormBound+UNION, using HR@5 as the evaluation metric. Across all defenses, StegaPoison consistently yields the lowest recommendation performance among

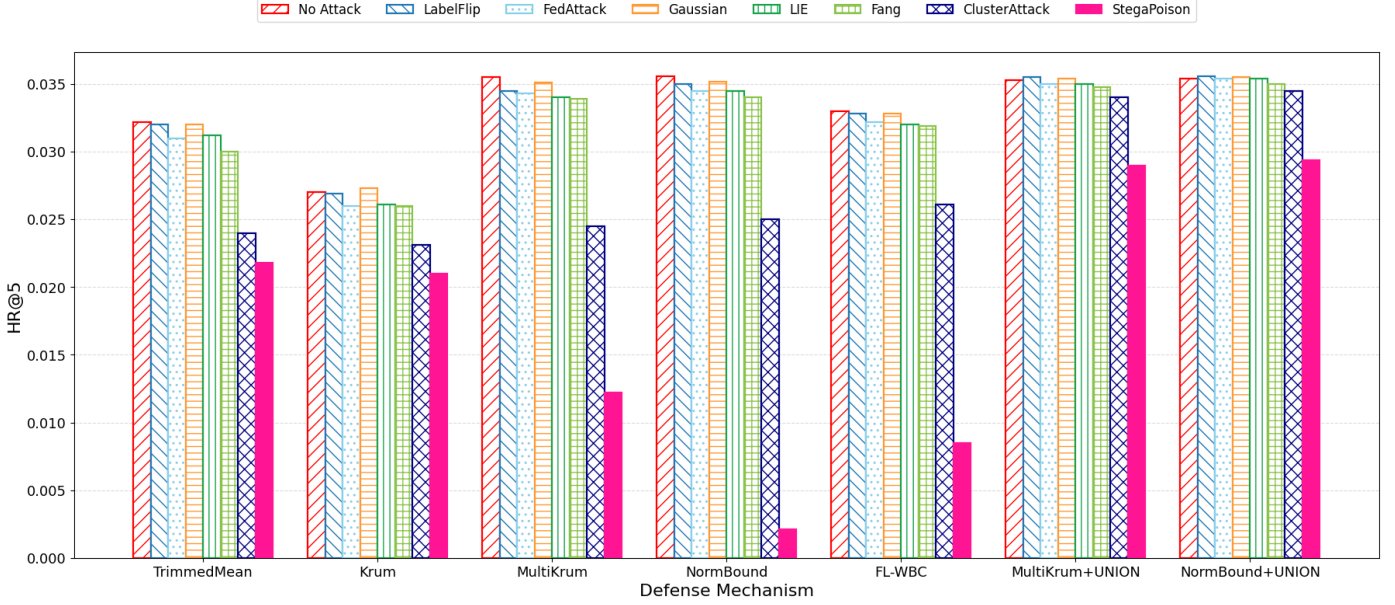


Figure 3: Attack Performance under Defense Mechanisms

all compared attacks, demonstrating that existing robust aggregation strategies remain ineffective against carefully crafted stealthy poisoning updates. The strongest degradation is observed under NormBound and FL-WBC, where HR@5 drops to approximately 0.002 and 0.009, respectively, indicating that norm-based filtering alone cannot mitigate the attack. Even under stronger hybrid defenses such as MultiKrum+UNION and NormBound+UNION, StegaPoison still reduces HR@5 to below 0.030, outperforming conventional baselines such as FedAttack, Gaussian, LIE, Fang, and ClusterAttack. In contrast, most existing attacks remain close to the benign “No Attack” baseline, suggesting that they are more easily neutralized by robust aggregation. Moreover, Fig. 4 shows the overall high performance and robustness of StegaPoison compared to ClusterAttack, which is our main baseline in this paper. These results confirm that StegaPoison successfully bypasses both distance-based and norm-based defenses by embedding malicious perturbations within statistically plausible client updates, thereby maintaining high attack effectiveness even in defended federated recommendation environments.

6.4 Stealthiness Analysis

In addition to achieving substantial degradation in recommendation quality, StegaPoison demonstrates significantly stronger stealth characteristics compared with existing untargeted poisoning attacks. Unlike conventional attacks such as Gaussian, LIE, Fang, and ClusterAttack, which often introduce statistically abnormal or highly concentrated gradient patterns, StegaPoison carefully embeds adversarial perturbations within naturally smooth embedding regions and constrains the poisoned updates to remain statistically consistent with benign client behavior. Specifically, the combination of low-variance dimension embedding perturbation (LVDEP), mirror-shift amplification, and statistical norm clipping enables the attack to preserve

the global distributional properties of benign gradients while still inducing severe semantic misalignment in the latent representation space. As a result, StegaPoison successfully bypasses robust aggregation defenses that rely on distance-based filtering, clustering consistency, or norm anomaly detection.

Furthermore, as illustrated in Fig. 5, while ClusterAttack attempts to evade defenses through coordinated malicious clustering, its updates still exhibit detectable aggregation patterns due to excessive directional similarity among malicious clients. In contrast, StegaPoison distributes perturbations more naturally across embedding dimensions and interaction dynamics, making the poisoned updates substantially harder to isolate from benign client updates. These observations indicate that embedding-space steganographic perturbation introduces a more covert and practically dangerous threat model for federated recommender systems.

6.5 Ablation Study of StegaPoison

To evaluate the contribution of each individual component of the proposed StegaPoison framework, we conduct an extensive ablation study across two benchmark datasets (MovieLens and Gowalla) using two recommendation architectures: Matrix Factorization (MF) and SASRec. We analyze eight distinct experimental configurations, ranging from isolated components to the unified attack. The performance impact is measured using Hit Ratio (HR@5) and Normalized Discounted Cumulative Gain (nDCG@5), as illustrated in Fig. 6a and Fig. 6b, respectively.

- **Exp-1 to Exp-4:** Individual components (Watermarking, LVDEP, Mirror Shift, and Velocity-Based Interaction Sampling) combined with Statistical Invisibility.
- **Exp-5 to Exp-7:** Various combinations of the aforementioned strategies as following: Exp-5 (Watermarking +

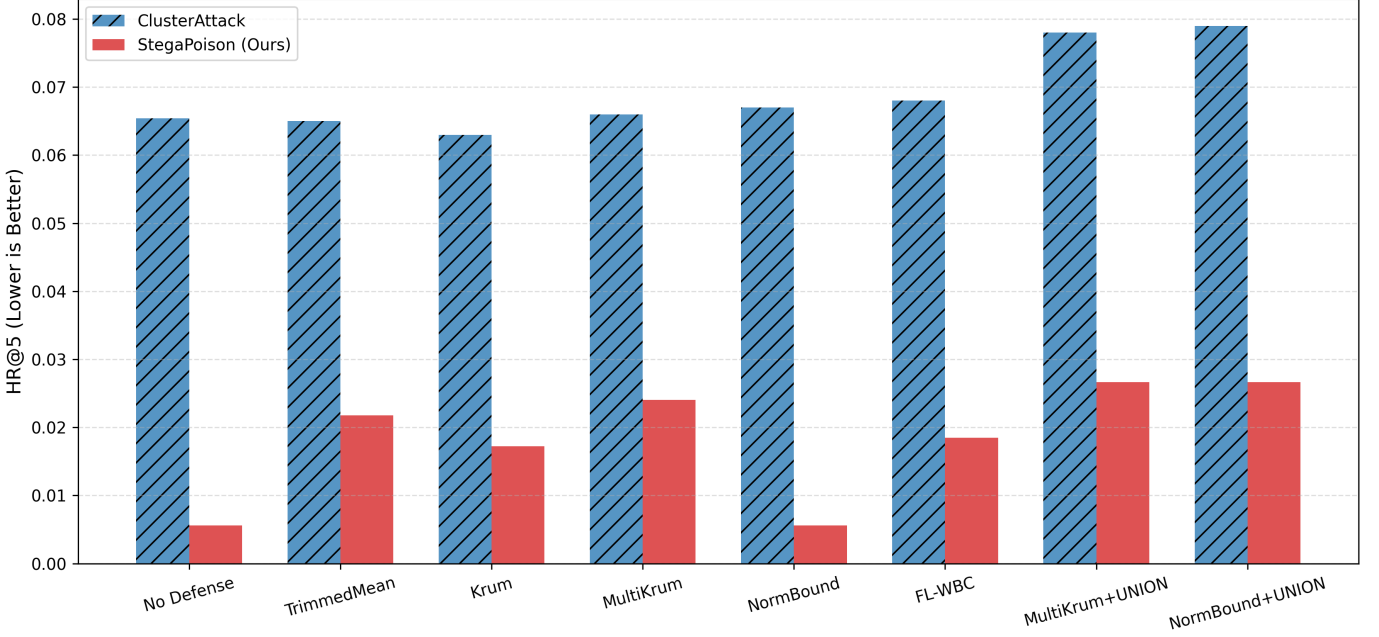


Figure 4: Defense Sensitivity Analysis - Attack Robustness Comparison

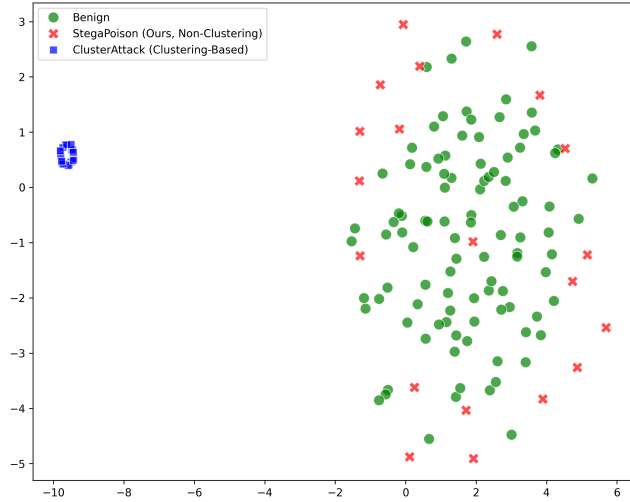


Figure 5: Gradient Distribution Analysis (t-SNE) - Non-Clustering Stealth

LVDEP + Statistical Invisibility), Exp-6 (Watermarking + LVDEP + Mirror Shift + Statistical Invisibility) and Exp-7 (Mirror Shift + Velocity Based Sampling + Statistical Invisibility).

- **Full Attack (Exp-8):** The integrated StegaPoison framework.

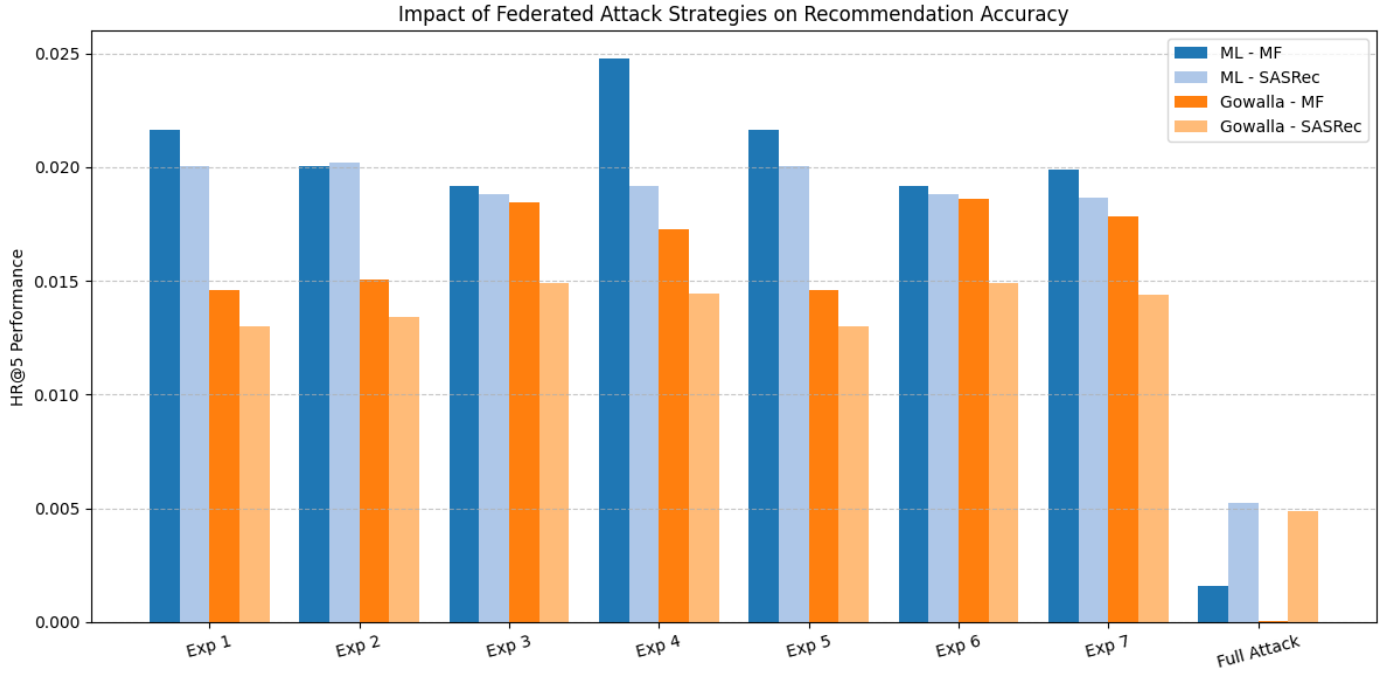
As shown in Fig. 6a, the individual components (Exp-1 through Exp-4) maintain relatively high HR@5 scores, indicating that while they introduce subtle perturbations, they do not possess sufficient disruptive power to collapse the recommendation utility on their own. For instance, in the MovieLens (ML-MF) scenario, these variants yield HR@5 values consistently above 0.015. Similarly, Fig. 6b demonstrates that the ranking qual-

ity (nDCG@5) remains stable across these partial implementations.

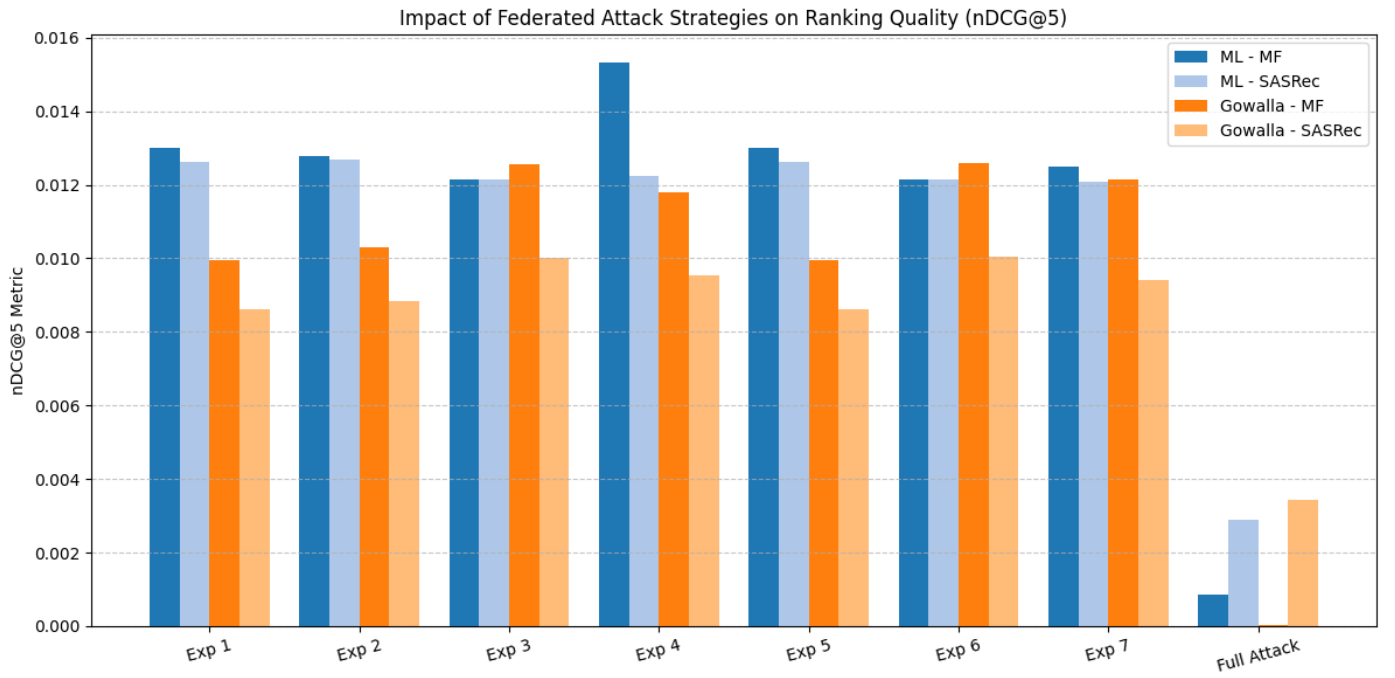
The experimental results reveal a significant performance drop—representing a successful attack—only when the components are fully integrated. In the “Full Attack” configuration, the recommendation accuracy is drastically neutralized. Specifically, in Fig. 6a, the HR@5 for Gowalla-MF drops to near-zero levels, and the ML-MF performance is reduced by over 90% compared to Exp-4. This trend is mirrored in Fig. 6b, where the ranking quality (nDCG@5) for the Full Attack exhibits a sharp decline across all datasets and models.

The ablation study confirms that StegaPoison’s superiority arises from synergistic interactions between its components, rather than any single dominant technique. We identify three key findings:

1. Dual-surface corruption is multiplicative, not additive. StegaPoison simultaneously poisons both user embeddings (via Watermarking and LVDEP) and item embeddings (via Mirror Shifting, Momentum (Velocity-Based Sampling), and Statistical Norm Clipping). Since the recommendation score is computed as a function of both embeddings, corrupting both surfaces amplifies the degradation beyond the sum of individual contributions. The Custom Ablation of Exp 7 (Mirror + Momentum + Stat. Invisibility, without any user-side perturbation) confirms this: despite activating the full item-side attack pipeline, it achieves results nearly identical to Exp 3 (Mirror alone), demonstrating that the item-side attack plateaus without concurrent user embedding corruption.
2. Momentum transforms one-shot perturbation into persistent structural damage. Without momentum accumulation (Exp 3, Exp 6), Mirror Shifting generates a different ran-



(a) Ablation Study - HR@5



(b) Ablation Study - nDCG@5

Figure 6: Ablation study results under different StegaPoison components.

dom destructive direction each round, leading to an undirected random walk that the model can partially recover from. With momentum, the attack direction is accumulated across rounds, ensuring that each round’s gradient contribution reinforces previous structural damage rather than canceling it. The comparison between Exp 6 (all techniques except Momentum) and the full attack isolates this effect.

3. Statistical Norm Clipping enables the destructive direction to survive aggregation. The norm clipping mechanism constrains the poisoned item gradients to lie within $\mu + k\sigma$ of the benign gradient norm distribution. This does not merely scale down the attack—it preserves the destructive direction established by Mirror Shifting while rendering the update statistically indistinguishable from honest clients. This synergy is precisely why StegaPoison maintains its effectiveness under robust aggregation defenses (e.g., Krum, TrimmedMean), where more aggressive attacks such as ClusterAttack are detected and filtered.

In summary, the full attack’s advantage is architectural: each component addresses a specific limitation of the others. Mirror Shifting provides the destructive direction, Momentum ensures persistence, Norm Clipping provides stealth, and LVDEP/Watermarking open a second attack surface on user embeddings. Removing any component breaks this chain, explaining the consistent performance gap observed across all ablation configurations.

6.6 Discussion

The empirical findings demonstrate that StegaPoison constitutes a highly effective untargeted poisoning threat against federated recommender systems. Across different backbone architectures, datasets, and evaluation metrics, the proposed method consistently causes substantial degradation in recommendation quality while preserving a low statistical profile. This behavior is particularly important because practical adversaries are not only interested in maximizing damage, but also in avoiding detection by existing aggregation or anomaly-filtering mechanisms. In this regard, the results suggest that embedding-space perturbations designed to mimic benign update characteristics can be more dangerous than conventional high-magnitude poisoning strategies.

A key implication of this work is that robustness in federated recommendation cannot be adequately assessed using only norm-based or distance-based defenses. Methods such as Krum, Trimmed Mean, and clipping mechanisms primarily assume that malicious updates are distinguishable through abnormal magnitude or directional inconsistency. However, StegaPoison exploits the fact that carefully structured perturbations may remain statistically similar to legitimate updates while still introducing severe semantic misalignment between user and item representations. This reveals an important vulnerability gap between *statistical anomaly detection* and *functional model integrity*. In other words, an update may appear normal numerically while being highly destructive behaviorally.

The strong performance of StegaPoison across both matrix factorization and sequential recommendation models also indicates that the attack is not tightly coupled to a single architecture. Although the internal parameterization of MF and SAS-Rec differs substantially, both rely on learned latent representations to model user preference signals. By perturbing these latent spaces in a covert manner, the attack transfers effectively across model families. This suggests that latent embedding manipulation may represent a broader attack surface in modern recommender systems, especially as representation learning becomes increasingly dominant.

Another noteworthy observation is the role of adaptive parameter selection through velocity-based sampling. Rather than perturbing parameters uniformly, StegaPoison prioritizes dimensions and interactions exhibiting rapid optimization dynamics. Moreover, StegaPoison adopts Adam optimizer (i.e. FedAdam) for this purpose. This strategy increases poisoning efficiency by focusing the attack budget on parameters that exert greater influence on convergence behavior. More generally, the results imply that temporal optimization signals can leak valuable information to adversaries regarding where malicious modifications will be most impactful.

From a defensive perspective, the study highlights the need for next-generation protection mechanisms for federated recommendation. Future defenses should move beyond simple magnitude filtering and instead consider representation consistency, cross-round behavioral drift, update provenance, and latent semantic alignment between user and item embeddings. Hybrid defenses combining robust aggregation with model-behavior auditing may offer stronger protection against stealth-oriented attacks such as StegaPoison.

Overall, StegaPoison demonstrates that stealthy manipulation of embedding updates can simultaneously achieve high attack utility and strong evasiveness. The broader lesson is that federated recommender systems require security mechanisms that reason not only about *how large* an update is, but also about *what the update means* in the latent preference space.

7. Limitations and Future Work

Although StegaPoison demonstrates strong attack effectiveness and high evasiveness across multiple experimental settings, several limitations should be acknowledged, which also motivate important future research directions. First, the current evaluation is conducted on widely used public benchmark datasets, namely MovieLens-1M and Gowalla. While these datasets enable standardized and reproducible comparisons, they may not fully reflect the scale, sparsity patterns, dynamic user behavior, and operational constraints of large industrial recommender systems. Future work will therefore investigate StegaPoison on larger and more realistic production-scale datasets with evolving item catalogs and continuously changing interaction patterns.

Second, this study focuses on two representative recommendation architectures, namely Matrix Factorization (MF) and

SASRec. Although these models capture both shallow latent-factor and sequential recommendation paradigms, modern recommender platforms increasingly employ graph neural networks, transformer-based models, multimodal recommenders, and hybrid retrieval-ranking pipelines. Extending the evaluation to such architectures would provide a broader understanding of the transferability and generalizability of StegaPoison across contemporary recommendation systems.

Third, the present attack formulation assumes a controlled proportion of malicious clients participating under relatively standard federated training conditions. In practical deployments, however, client participation may be intermittent, sampling may be highly dynamic, and adversarial availability may vary over time. Accordingly, future work should explore more realistic federated environments characterized by asynchronous communication, partial participation, non-IID data distributions, and dynamic adversarial behavior.

Fourth, the current framework concentrates on untargeted degradation of overall recommendation quality rather than targeted objectives. While untargeted attacks are valuable for evaluating system robustness, targeted scenarios such as item promotion, item suppression, and group-specific manipulation may better capture incentives in commercial recommendation ecosystems. Investigating these objectives would further broaden the threat landscape addressed by StegaPoison.

Fifth, although the proposed method is evaluated against several representative robust aggregation and anomaly-based defenses, it does not exhaustively cover all recent protection strategies. Future studies should benchmark StegaPoison against more advanced defenses, including representation-aware anomaly detection, trust-based aggregation, certified robustness methods, blockchain-assisted auditing, and secure execution frameworks.

Finally, the current analysis does not fully quantify the trade-offs among stealthiness, attack strength, computational overhead, and communication efficiency under all hyperparameter settings. A systematic study of these trade-offs would provide deeper insight into both attack optimization and the design of more effective defenses.

Despite these limitations, the results consistently indicate that covert embedding-space poisoning remains a serious and underexplored threat to federated recommender systems, motivating the development of scalable and semantically aware security mechanisms.

8. Conclusion

This paper presented StegaPoison, a novel steganography-inspired untargeted poisoning attack against federated recommender systems. Unlike conventional poisoning methods that often rely on large or easily distinguishable malicious updates, StegaPoison introduces carefully structured perturbations that remain statistically benign while substantially degrading recommendation performance. The proposed framework combines sinusoidal watermark-style modulation, low-variance di-

mension embedding perturbation (LVDEP), mirror-shift manipulation of item embeddings, velocity-based parameter selection, and norm-aware concealment to generate highly effective yet stealthy poisoned updates.

Extensive experiments on representative recommendation benchmarks, including MovieLens-1M and Gowalla, demonstrated that StegaPoison consistently outperforms existing untargeted attack baselines under multiple settings. The attack achieved severe reductions in HR@5 and nDCG@5 across both matrix factorization and sequential recommendation architectures, indicating strong cross-model generalizability. Furthermore, the results under robust aggregation schemes such as Krum, Trimmed Mean, Multi-Krum, and clipping-based defenses showed that many current protection mechanisms remain vulnerable when malicious updates are statistically camouflaged rather than overtly anomalous.

Beyond the empirical results, this work highlights an important security insight for federated recommendation: preserving numerical consistency of client updates does not necessarily preserve model integrity. Malicious perturbations can be embedded within normal-looking embedding updates while still inducing substantial semantic misalignment in latent user-item representations. Therefore, future secure federated recommender systems should incorporate defenses that monitor not only update magnitude and direction, but also behavioral and representational consistency over time.

In summary, StegaPoison demonstrates that steganography-inspired manipulation constitutes a realistic and powerful threat to federated recommender systems. We hope this work motivates the development of stronger security benchmarks and more semantically aware defenses for next-generation federated recommendation platforms.

CRedit authorship contribution statement

Shakhawan Mahmood: Conceptualization, Data Curation, Formal analysis, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing; **Sabah Robitan:** Conceptualization, Data curation, Formal analysis, Software, Visualization, Writing – review & editing; **Beste Ustubioglu:** Conceptualization, Formal analysis, Writing – review & editing; **Gul Tahaoglu:** Conceptualization, Formal analysis, Writing – review & editing; **Arda Ustubioglu:** Conceptualization, Formal analysis, Writing – review & editing; **Guzin Ulutas:** Conceptualization, Investigation, Supervision, Project administration, Formal analysis, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing

financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT and Gemini in order to reorganize the ideas, concepts, proofreading, and error checking. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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